Practical - 5

Registration Form with Frontend Validation and User-Friendly Error Handling

A . Theory

Create a student registration form with HTML5 input types and JavaScript validation for name, email, mobile number, password, confirm password, course, year, gender, and terms acceptance. Use Regular Expression for validations.

1. **Are correct input types and attributes used?**
   Yes. We use suitable types such as `text`, `email`, `password`, `tel`, and `date`, along with attributes like `required`, `pattern`, `minlength`, and `maxlength`.
2. **Are validation errors displayed near the relevant fields?**
   Yes. Validation error messages are displayed next to or below the field where the error occurs, so the user can easily identify and correct it.
3. **Is password strength checked?**
   Yes. Password strength can be checked using conditions such as minimum length and the presence of uppercase letters, lowercase letters, numbers, and special characters.
4. **Is the form accessible using keyboard and screen-reader-friendly labels?**
   Yes. Each input has a proper `<label>` connected using the `for` and `id` attributes. The form can also be navigated using the keyboard with the **Tab** key.

B. Implementation/ Output/Result

Here is my GitHub repo link the codes are here in repo:
https://github.com/dhyaani19/StudentHub-Portal.git

C.. Evidence of Work (Screenshots)

Figure 5.1 : Profile page (peofile.html)

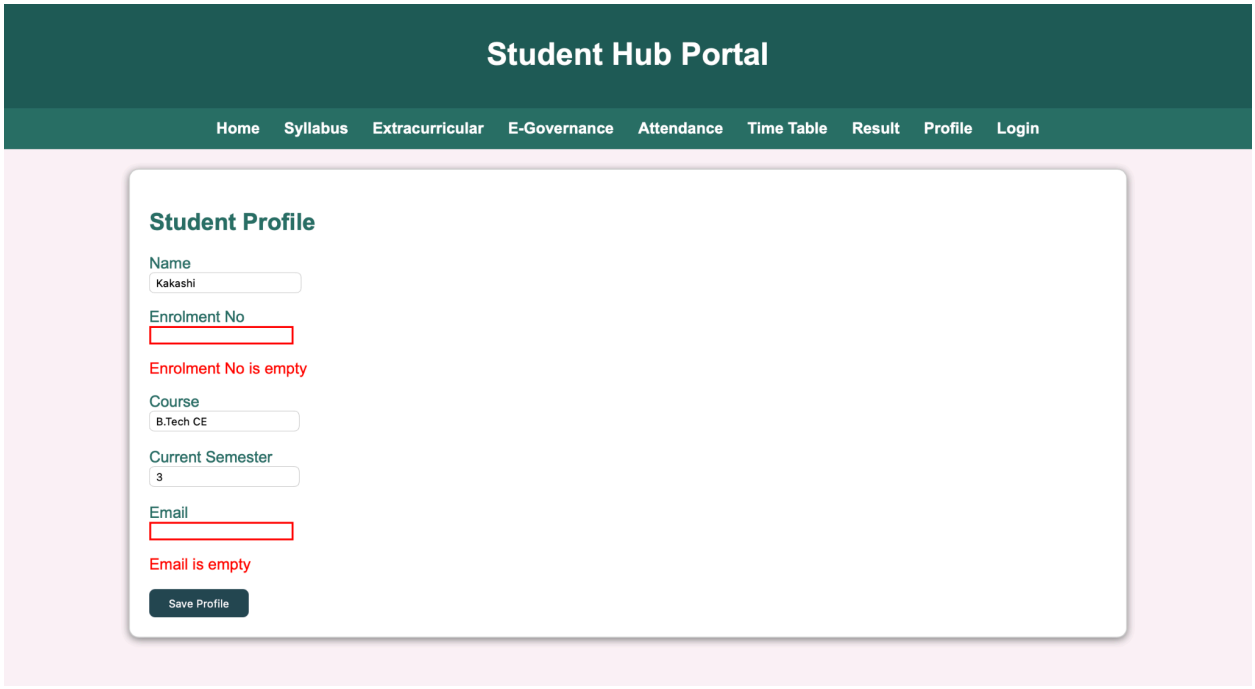


Figure 5.2 : Profile page (peofile.html)

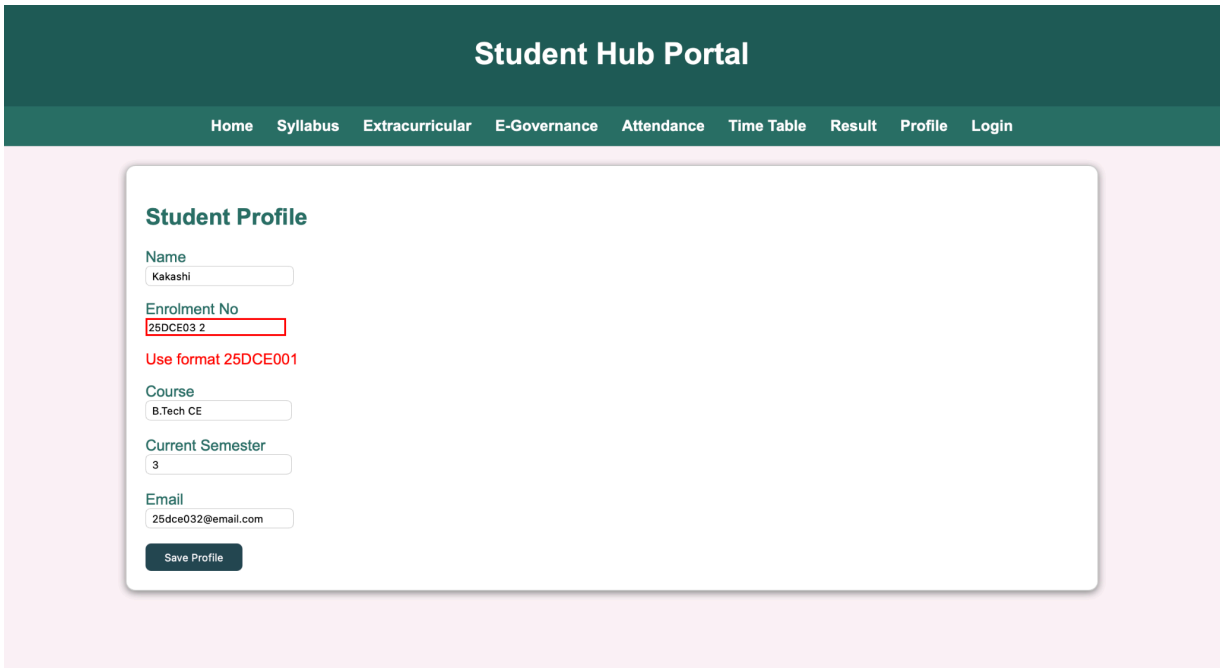


Figure 5.3 : Profile page (peofile.html)

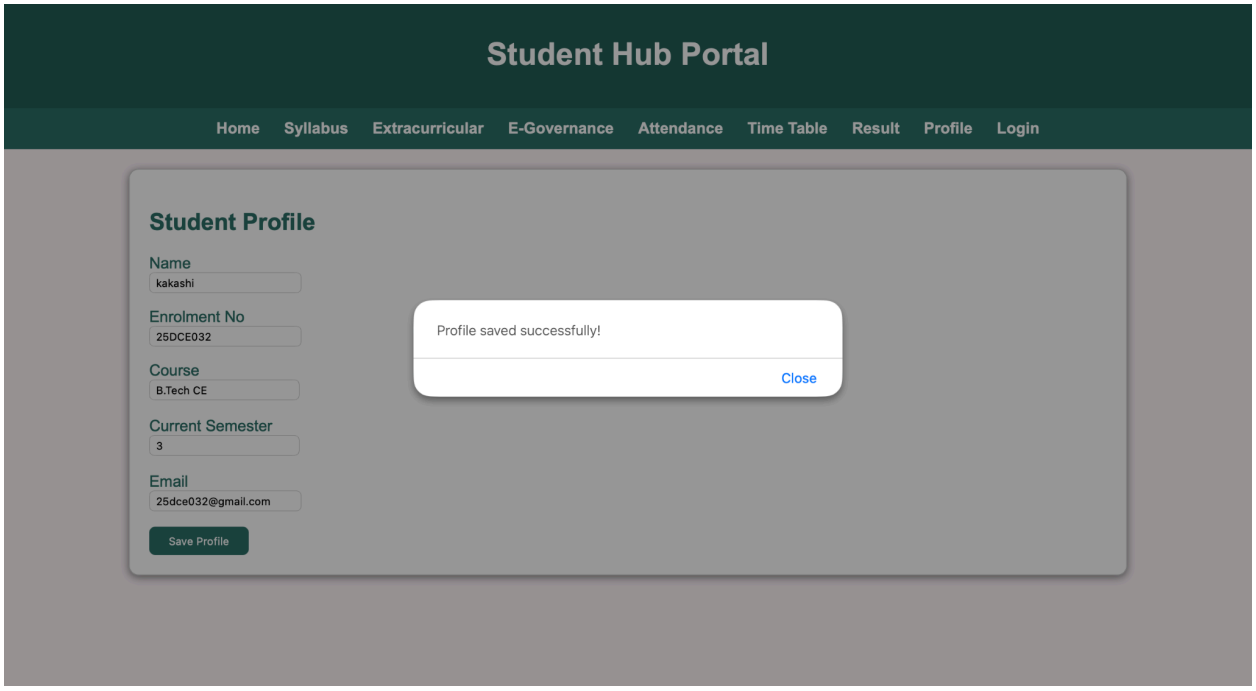


Figure 5.4 : Login page (login.html)

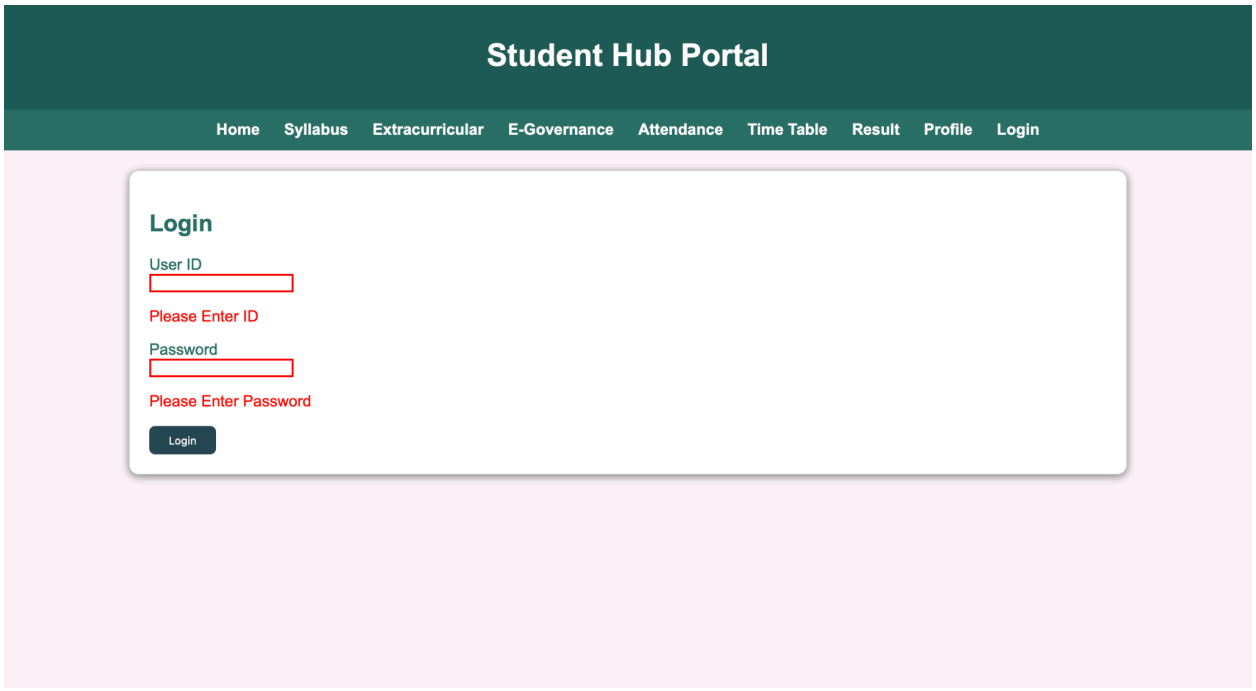
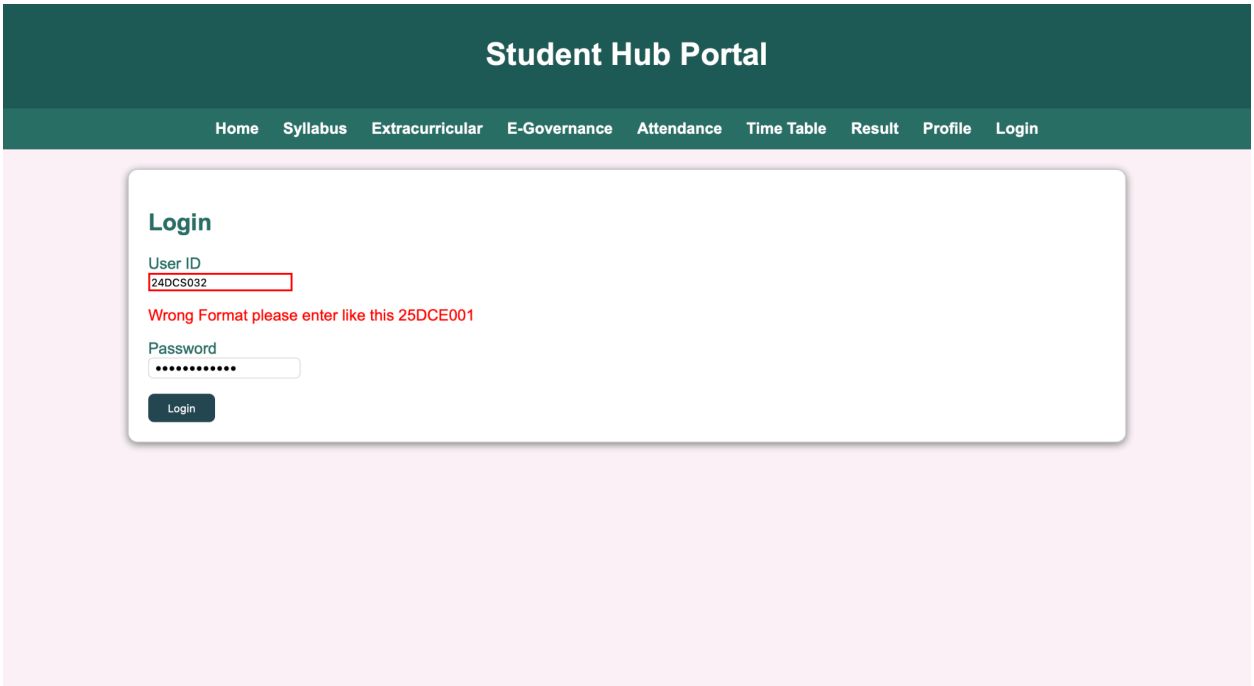


Figure 5.5 :Login page (login.html)



D. GitHub Repository Link:

https://github.com/dhyaani19/StudentHub-Portal.git

E. Challenges Faced:

While doing this practical, I faced some difficulties in understanding how to validate user input using JavaScript. Creating the correct Regex for different fields was a little confusing at first. I also had some problems showing the error message at the correct place and changing the input box style when the data was invalid. Sometimes the validation worked for one field but not for another. I solved these problems by checking my code carefully and testing each input field one by one**.**

F. Learning Outcomes:

n this practical, I learned how to take input from the user using JavaScript and validate the entered data. I learned how to check if a box is empty and show an error message

when the input is invalid. I also learned how to use Regular Expressions (Regex) to validate different fields in the Login and Profile forms. I learned how to change the input box style when the entered data is incorrect. Overall, I understood how JavaScript can be used to make forms more interactive and user-friendly.

Faculty Remarks: